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Theory of Sound Propagation in a Duct with a Branched Tube Using Modal Decomposition

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Summary
The present paper extends the scope of a previous paper of one of the authors [1]. The modal decomposition is used to obtain both an exact result in a matrix form and a variational formulation for the description of the insertion of a branched tube on a main waveguide where only the planar mode propagates. Two representations are given: a two-port junction and a three-port junction, with their equivalent circuits. Results are given for the 2D, rectangular geometry case, with a discussion of some approximations and convergence problems. Concerning the circular geometry, new formulae are given and compared with results determined with the boundary element method.

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1. Introduction

The problem of a tube branched on a waveguide has been investigated by many authors for various applications: e.g. ventilation ducts, branched resonators, woodwind musical instruments. In this study the axes of the main and side duct are perpendicular (Figure 1).

Fifteen years ago, one of the authors [1, 2] published a study of the problem based on modal decomposition and variational calculation. He adapted methods developed for microwave circuits [3] and gave for the first time formulae for the cylindrical geometry. The derivation is essentially correct, but contains some errors: it is the first aim of the present paper to rewrite the theory and give some new results, obtained by Dubos [4], which differ slightly from Keefe's first results. These results are presented in the last section (5) with a discussion about their validity: it will be shown that the cylindrical geometry is not well adapted to the modal decomposition, but with reasonable assumptions it is possible to obtain good approximations. As a matter of fact these approximations have been compared to other calculations, especially those in a recent paper using a finite difference method by Nederveen et al [5]. Other calculations based upon the finite element method are presented here. Comparison with experiments can be found in the thesis report by Dubos [4] and will be published later.

Our paper will present some complements to the derivation of the final results concerning the cylindrical geometry:

i) In section 2, we treat the question of symmetry. We formulate the case of acoustic wave propagation in a duct, which is represented as a transmission line, on which a branched tube is placed. We make no assumption regarding the plane of symmetry of the branched tube with respect to the origin of the transmission line, and we model the acoustic effect of the branched tube using an equivalent circuit. The intrinsic limitations of the classical T- or Π-shaped circuit are discussed.

ii) In section 3 a matrix formalism is used to obtain a modal decomposition of the velocity field in the branched tube (see [6]). The variational formulation used by Keefe [1, 2] is presented in a general context. Moreover, if the theory is given first for a side tube assumed to have a known input impedance, or a known Green function, then it is given also for a side tube playing the same rôle as the main duct: it means that a three-port junction is considered instead of a two-port junction. Each port corresponds to the planar mode in each duct because the present study is limited to the low frequency range. (A general presentation of the junctions between several ducts can be found in [7]).

iii) In section 4, the two-dimensional (2D) case is treated, for which the results at zero frequency are well known (using the conformal representation [8], or [9, 10] in the acoustical context). This case is useful for testing the validity of the modal decomposition method as well as several other approximations and calculation methods. The modal decomposition in the 2D-case has often been studied [11, 12, 13, 14, 15]. Here, we specially present the plane piston approximation and compare it to the exact result, and discuss several questions of convergence, already treated for a step discontinuity [16]. Finally we present briefly some results for higher frequencies, below the first cutoff of the main guide.

iv) Section 5 treats the case of tubes with circular cross-sections. This case is specially difficult because of the saddle

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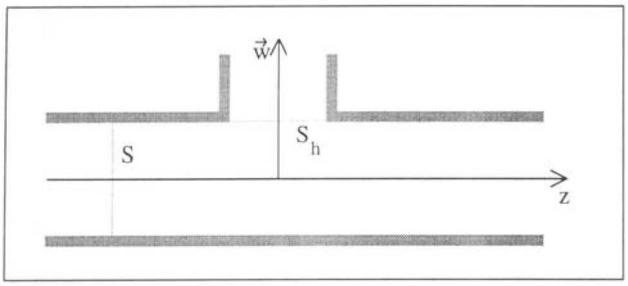

Figure 1. Schematic view of the general problem.

shape of the matching surface between the branched tube and the main waveguide. Nevertheless, analytical formulae were calculated using several approximations, showing that the values given by Keefe in his first study were not too bad, also taking into account several uncertainties. Finally the results are compared with those of the Boundary Element Method, and illustrate the great difficulty of the use of such methods for the study of discontinuities at low frequencies.

2. Transmission line representation of a branched tube

2.1. Basic integral equation

Consider an infinite air column of uniform cross-section S oriented along the z-axis (see Figure 1). The frequency is assumed to be sufficiently low so that only the planar mode propagates, whereas the remainder of the modes are evanescent. The midpoint of the branched tube is located at $z = 0$.

In the absence of the branched tube discontinuity, the pressure p_f associated with the propagating mode in the air column depends only on the axial z-coordinate, and the general solution to the wave equation may be decomposed into two components, one travelling to the right (+) and one to the left (−):

$$p_f(\boldsymbol{r}) = p_f^+ \mathrm{e}^{-\mathrm{j}kz} + p_f^- \mathrm{e}^{+\mathrm{j}kz}, \qquad (1)$$

where k is the wavenumber, $p_f^\pm$ are two complex coefficients, and $\boldsymbol{r} = (\boldsymbol{w}, z)$ is the spatial coordinate in the main air column, the transverse section being defined by the two-dimensional coordinate $\boldsymbol{w}$. A harmonic time dependence $\mathrm{e}^{\mathrm{j}\omega\tau}$ is assumed throughout, where $\mathrm{j} = \sqrt{-1}$, ω is the angular frequency and τ the time.

The presence of the branched tube modifies the air column pressure, and the pressure $p_f(\boldsymbol{r})$ at any point in the air column can be written in terms of $p_f(\boldsymbol{r})$ and an integral over the surface of the infinite main waveguide, according to the Helmholtz-Huyghens equation [17, 18]:

$$p(\boldsymbol{r}) = p_f(\boldsymbol{r}) + \int_{\substack{\text{surface of the}\\ \text{infinite guide}}} \Big[G(\boldsymbol{r},\boldsymbol{r}')\partial_n p(\boldsymbol{r}') - p(\boldsymbol{r}')\partial_n G(\boldsymbol{r},\boldsymbol{r}')\Big]\,\mathrm{d}S', \qquad (2)$$

where ∂_n is the outward normal derivative to the surface, and G is the Green function for the main air column.

The following assertions simplify this equation by restricting the integration over the surface of intersection S_h between the main waveguide and the branched tube. The integrals over both ends of the infinite main waveguide, which represent the pressure generated on these surfaces, vanish when the Sommerfeld condition is applied, because there is no source at infinity. Moreover, we choose the Green function for the main waveguide to satisfy the Neumann conditions (i.e., its normal derivative vanishes on the surface of the waveguide). Finally the derivative of the pressure vanishes on the walls of the main air column. Therefore equation (1) can be written as the sum of the pressure in the absence of the branched tube and an integral over the surface of intersection between the main waveguide and the branched tube:

$$p(\boldsymbol{r}) = p_f(\boldsymbol{r}) + \int_{S_h} G(\boldsymbol{r},\boldsymbol{r}')\partial_n p(\boldsymbol{r}')\,\mathrm{d}S'. \qquad (3)$$

Another equivalent explanation of this equation can be found in [6].

Then neglecting dissipation and according to the Euler equation, equation (3) becomes:

$$p(\boldsymbol{r}) = p_f(\boldsymbol{r}) - \mathrm{j}\omega\rho \int_{S_h} G(\boldsymbol{r},\boldsymbol{r}')v(\boldsymbol{r}')\,\mathrm{d}S', \qquad (4)$$

where v is the particle velocity component normal to the surface of intersection, and ρ is the equilibrium density of air. The equation is valid for every choice of pressure $p_f(\boldsymbol{r})$, but our interest is the pressure field at sufficiently low frequencies so that only the fundamental mode propagates, and all higher order modes are evanescent. Thus, this pressure field $p_f(\boldsymbol{r})$ depends on the axial coordinate z only.

2.2. The Green function for the infinite main waveguide

The Green function for the infinite main waveguide may be represented in terms of a sum over the response of the plane-wave mode with eigenvalue k, and the evanescent eigenmodes $\psi_{mn}(\boldsymbol{w})$ with corresponding eigenvalues k_{mn} [17, 18] by:

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{\mathrm{e}^{-\mathrm{j}k|z-z'|}}{2\mathrm{j}kS} + \sum_{m+n>0}^{\infty} \frac{\mathrm{e}^{-\mathrm{j}k_{mn}|z-z'|}}{2\mathrm{j}k_{mn}S}\psi_{mn}(\boldsymbol{w})\psi_{mn}(\boldsymbol{w}'). \qquad (5)$$

The eigenfunctions ψ_{mn} satisfy the homogeneous Helmholtz equation:

$$\Delta\psi_{mn} + \eta_{mn}^2\psi_{mn} = 0, \qquad (6)$$

and

$$k_{mn}^2 = k^2 - \eta_{mn}^2,$$

and the following orthogonality condition:

$$\int_S \psi_{im}(\boldsymbol{w})\psi_{jn}(\boldsymbol{w})\,\mathrm{d}S = S\delta_{ij}\delta_{mn}, \qquad (7)$$

where δ_{ij} is the Kronecker symbol (we have chosen the eigenfunctions to be dimensionless).

2.3. Planar mode in the main waveguide

Outside the entrance surface of the branched tube S_h, the pressure field can also be expanded on the modes of the main waveguide:

$$p(\boldsymbol{r}) = \sum_{m,n} \psi_{mn}(\boldsymbol{w}) p_{mn}(z). \tag{8}$$

Then, projecting the integral equation (4) on the planar mode $p_{00}(z)$, noted $p_0(z)$ the following result is obtained:

$$p_0(z) = p_f(z) - \frac{1}{2}\frac{\rho c}{S}\int_{S_h} \mathrm{e}^{-\mathrm{j}k|z-z'|} v(\boldsymbol{r}')\,\mathrm{d}S' \tag{9}$$

(the subscript 0 is not necessary for the field p_f, because it is assumed to be planar). This equation shows that the field $p_0(z)$ depends on the higher order modes only through the velocity over the surface S_h.

Instead of equation (1), the decomposition of the pressure field p_f into a symmetrical component (subscript s) and an antisymmetrical one (subscript a) with respect to an arbitrary origin, will be useful in the following:

$$p_f = p_{fs}\cos(kz) + p_{fa}\mathrm{j}\sin(kz). \tag{10}$$

Similarly, for the field $p_0(z)$, one can make two decompositions, valid either to the left (subscript L) or to the right (subscript R) of the branched tube:

$$\begin{cases} p_0(z) = p_L\cos(kz) - Z_c u_L \mathrm{j}\sin(kz), \\ p_0(z) = p_R\cos(kz) - Z_c u_R \mathrm{j}\sin(kz), \end{cases} \tag{11}$$

where u_L and u_R have the dimension of volume velocity, and $Z_c = \rho c/S$ is the characteristic impedance of the main waveguide. From equation (9), one obtains, by identification, taking into account the sign of the difference $(z - z')$:

$$\left|\begin{aligned} p_L &= p_{fs} - \tfrac{1}{2}Z_c(K_s - \mathrm{j}K_a), \\ p_R &= p_{fs} - \tfrac{1}{2}Z_c(K_s + \mathrm{j}K_a), \\ -u_L &= Z_c^{-1}p_{fa} - \tfrac{1}{2}(K_s - \mathrm{j}K_a), \\ -u_R &= Z_c^{-1}p_{fa} + \tfrac{1}{2}(K_s + \mathrm{j}K_a), \end{aligned}\right. \tag{12}$$

where

$$\left|\begin{aligned} K_s &= \int_{S_h} \cos(kz)v(\boldsymbol{r})\,\mathrm{d}S, \\ K_a &= \int_{S_h} \sin(kz)v(\boldsymbol{r})\,\mathrm{d}S. \end{aligned}\right. \tag{13}$$

If the velocity profile at the input of the branched tube is known, the problem is solved when K_s and K_a are known. By rewriting equations (12), the following equations are obtained:

$$\begin{cases} p_L - p_R = \mathrm{j}Z_c K_a, \\ u_L - u_R = K_s, \end{cases} \tag{14}$$

and

$$\begin{cases} p_L + p_R &= 2p_{fs} - Z_c K_s, \\ Z_c(u_L + u_R) &= -2p_{fa} - Z_c\mathrm{j}K_a. \end{cases} \tag{15}$$

These equations do not fix any origin. Moreover, equations (11) can be used even if the abscissa $z = 0$ is located on the surface S_h: in that case, the decomposition into a planar mode and higher order modes near $z = 0$ is fictitious, but the use of these equations at $z = 0$ is very convenient in order to calculate the transfer matrices between the abscissa $z = 0$ and a given abscissa z. The branched tube therefore transforms the quantities (p_L, u_L) into (p_R, u_R) via a transfer matrix. This idea was developed by Schwinger [3]. Equations (14) lead to this transfer matrix, if one knows the velocity $v(\boldsymbol{r})$ from an integral equation on the surface S_h expressing the input condition for the branched tube (see section 2.4).

From equations (15), we first derive an equation useful in the following:

$$\begin{aligned} 2p_f &= \cos(kz)(p_l + p_R) \\ &\quad - \mathrm{j}\sin(kz)Z_c(u_l + u_R) + T_f, \end{aligned} \tag{16}$$

where $T_f = Z_c[K_s\cos(kz) + K_a\sin(kz)]$, which can be rewritten as

$$\begin{aligned} T_f &= Z_c\int_{S_h} \cos\left[k(z - z')\right]v(\boldsymbol{r}')\,\mathrm{d}S' \\ &= -2\omega\rho\int_{S_h} \Im m\left\{G(z, z')\right\}v(\boldsymbol{r}')\,\mathrm{d}S'. \end{aligned} \tag{17}$$

As a matter of fact, the imaginary part of the Green function includes only one term, related to the planar mode (more generally, only the propagating modes contribute to the imaginary part).

Finally, if the branched tube is symmetrical with respect to a plane perpendicular to the z-axis, the velocity field $v(\boldsymbol{r})$ on the surface S_h can be decomposed into a symmetrical part, $v_s(\boldsymbol{r})$, and an antisymmetrical part, $v_a(\boldsymbol{r})$. Using this plane as the origin for the z-axis, the equations (13) can be simplified as follows:

$$\left|\begin{aligned} K_s &= \int_{S_h} \cos(kz)v_s(\boldsymbol{r})\,\mathrm{d}S, \\ K_a &= \int_{S_h} \sin(kz)v_a(\boldsymbol{r})\,\mathrm{d}S. \end{aligned}\right. \tag{18}$$

2.4. Input condition for the branched tube

At the input of the branched tube, an equation similar to equation (4) can be written for locations r within the branched tube:

$$p(\boldsymbol{r}) = \mathrm{j}\omega\rho\int_{S_h} G_h(\boldsymbol{r}, \boldsymbol{r}')v(\boldsymbol{r}')\,\mathrm{d}S'. \tag{19}$$

The difference in sign comes from the unique symbol for the normal velocity $v(\boldsymbol{r}')$. The Green function G_h is chosen to satisfy:

i. $\partial_n G_h = 0$ on the surface S_h,
ii. the same condition as $p(\boldsymbol{r})$ at the output of the side tube: for example, a radiation condition if it is open, or condition i. if it is closed.

2.5. Integral equation for the velocity on surface S_h

Equations (4) and (19) lead to the integral equation giving the velocity normal to the surface S_h with respect to the field:

$$\mathrm{j}\omega\rho \int_{S_h} \Big[G_h(\boldsymbol{r},\boldsymbol{r}') + G(\boldsymbol{r},\boldsymbol{r}')\Big] v(\boldsymbol{r}')\,\mathrm{d}S' = p_f(z). \quad (20)$$

We point out that the second member depends on the coordinate z only. Using equations (16) and (17), equation (20) can be modified as follows:

$$\mathrm{j}\omega\rho \int_{S_h} \Big[G_h(\boldsymbol{r},\boldsymbol{r}') + \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\Big] v(\boldsymbol{r}')\,\mathrm{d}S' = \frac{1}{2}\cos(kz)(p_L + p_R) - \frac{1}{2} Z_c \mathrm{j}\sin(kz)(u_L + u_R). \quad (21)$$

Thanks to this formulation, the imaginary part of the Green function of the main waveguide is eliminated. The importance of this is evident because the imaginary part diverges when the frequency tends to zero. Equations (21), (14) and (13) allow us to solve the matching problem between the planar modes (p_L, u_L) and (p_R, u_R).

From equation (21), we deduce that the velocity $v(\boldsymbol{r})$ is a combination of the quantities $(p_L + p_R)$ and $(u_L + u_R)$: it is therefore the same for the integrals K_s and K_a. For simplicity, we define two dimensionless components of the velocity:

$$v(\boldsymbol{r}) = w_1(\boldsymbol{r})\frac{p_L + p_R}{\rho c} + w_2(\boldsymbol{r})\frac{u_L + u_R}{\rho c} Z_c, \quad (22)$$

and equations (21) can be divided into the two following equations:

$$\left| \begin{aligned} &2\mathrm{j}k \int_{S_h} \Big[G_h(\boldsymbol{r},\boldsymbol{r}') + \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\Big] w_1(\boldsymbol{r}')\,\mathrm{d}S' = \cos(kz), \\ &2\mathrm{j}k \int_{S_h} \Big[G_h(\boldsymbol{r},\boldsymbol{r}') + \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\Big] w_2(\boldsymbol{r}')\,\mathrm{d}S' = -\mathrm{j}\sin(kz). \end{aligned} \right. \quad (23)$$

For the case where the branched tube has a symmetry plane at $z = 0$, equation (21) shows that the term w_1 is proportional to the symmetrical component of the velocity v_s (then $u_L = -u_R$) and the term w_2 is proportional to the antisymmetrical component of the velocity v_a (then $p_L = -p_R$).

2.6. Transfer matrix and equivalent circuits

We are now able to calculate the transfer matrix $\boldsymbol{T}$ between the planar mode on both sides of the branched tube (see equations 11). Actually the simplest relationship is not the classical transfer matrix, but an equivalent matrix. From equations (13), (14) and (21) this is found to be:

$$\left| \begin{aligned} p_L - p_R &= T_1(p_L + p_R) + T_2(u_L + u_R), \\ u_L - u_R &= T_3(p_L + p_R) + T_4(u_L + u_R), \end{aligned} \right. \quad (24)$$

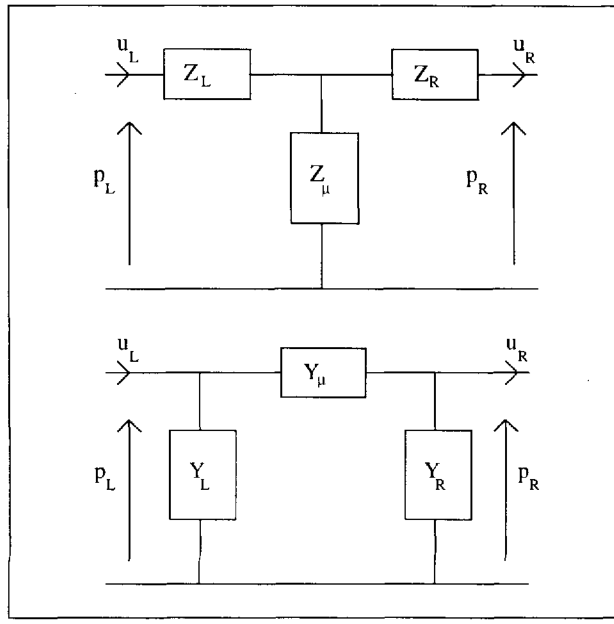

Figure 2. T-shaped and Π-shaped circuits for the 2-port network linking the left- and right-hand sides of the branched tube (see equations 27 and 28).

where

$$\begin{aligned} T_1 &= \mathrm{j}\frac{1}{S}\int_{S_h} \sin(kz) w_1(\boldsymbol{r}')\,\mathrm{d}S, \\ T_2 &= \mathrm{j}\frac{Z_c}{S}\int_{S_h} \sin(kz) w_2(\boldsymbol{r}')\,\mathrm{d}S, \\ T_3 &= \frac{1}{SZ_c}\int_{S_h} \cos(kz) w_1(\boldsymbol{r}')\,\mathrm{d}S, \\ T_4 &= \frac{1}{S}\int_{S_h} \cos(kz) w_2(\boldsymbol{r}')\,\mathrm{d}S. \end{aligned} \quad (25)$$

Using equations (23) and (25), reciprocity can be verified:

$$T_1 = -\frac{2\mathrm{j}k}{S}\int_{S_h}\int_{S_h} \Big[G_h(\boldsymbol{r},\boldsymbol{r}') + \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\Big] \cdot w_2(\boldsymbol{r}) w_1(\boldsymbol{r}')\,\mathrm{d}S\,\mathrm{d}S' = -T_4. \quad (26)$$

The matrix expression involved in equations (24) is the simplest relation for the two- port network we are expecting. We emphasize the fact that, in equations (23), if losses are ignored and the branched tube is finite, the Green function G_h is real, so w_1 and w_2 are real and purely imaginary quantities, respectively. We can deduce other matrices, such as transfer, scattering, impedance or admittance matrix, or equivalent circuits (T-shaped or Π-shaped). It appears that each of these representations of the two-port network is more complicated than the above expressions (24). The elements of the T-equivalent circuit are (see Figure 2):

$$\left| \begin{aligned} Z_L &= T_2 - \frac{T_4 + T_1 T_4}{T_3}, \\ Z_R &= T_2 + \frac{T_4 - T_1 T_4}{T_3}, \\ Z_\mu &= \frac{1}{2}\Big(-T_2 + \frac{1 + T_1 T_4}{T_3}\Big). \end{aligned} \right. \quad (27)$$

The admittance matrix is obtained from the impedance matrix by inverting T_3 and T_2, and T_1 and T_4, and the Π-shaped circuit (see Figure 2) is obtained similarly from the T-shaped circuit. If the plane $z = 0$ is a plane of symmetry for the branched tube, we have the property $T_1 = T_4 = 0$. Thus the two circuits are greatly simplified:

$$\begin{aligned} Z_L &= Z_R = T_2, & Z_\mu &= \frac{1}{2}\Big(- T_2 + \frac{1}{T_3}\Big), \\ Y_L &= Y_R = T_3, & Y_\mu &= \frac{1}{2}\Big(- T_3 + \frac{1}{T_2}\Big). \end{aligned} \tag{28}$$

$Z_L = Z_R$ corresponds to the antisymmetrical component of the velocity only (similarly $Y_L = Y_R$ to the symmetrical one). Z_μ and Y_μ involve two components; this is certainly a disadvantage of this representation.

3. Solutions of the integral equations by using modal decomposition of the velocity on surface S_h

3.1. Statement of the problem

The integral equations (23) can be solved by using several techniques [19, 20]. A natural solution is the decomposition of the velocity $v(\boldsymbol{r})$, and therefore of the quantities $w_1(\boldsymbol{r})$ and $w_2(\boldsymbol{r})$, on the modes of the side branch. It is only possible if the surface S_h is perpendicular to the side tube, especially if the main tube cross section is rectangular, so that on the surface S_h, the velocity $v(\boldsymbol{r})$ does not depend on the longitudinal coordinate of the side branch.

For this decomposition, we use a matrix formalism [6]. Column vectors $\varphi(\boldsymbol{r})$, $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are defined, their elements being the mode eigenfunctions $\varphi_i(\boldsymbol{r})$, which depend on the transverse coordinates of the side branch (we order them for having a single infinity of modes), and the modal coefficients of w_1 and w_2 on S_h , respectively. The modal decomposition is written on the surface S_h as:

$$\begin{aligned} w_1(\boldsymbol{r}) &= \frac{\rho c}{2S_h}\, {}^t\varphi(\boldsymbol{r})\boldsymbol{W}_1, \\ w_2(\boldsymbol{r}) &= \frac{\rho c}{2S_h}\, {}^t\varphi(\boldsymbol{r})\boldsymbol{W}_2, \end{aligned} \tag{29}$$

where the superscript t indicates the transpose of a vector (or a matrix) and the factor $\rho c/2S_h$ is convenient for the following calculations. The elements of the vectors $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ have the dimension of an admittance, and the modes φ_i are normalized by:

$$\int_{S_h} \varphi(\boldsymbol{r})\, {}^t\varphi(\boldsymbol{r})\, \mathrm{d}S = S_h \mathbf{1}, \tag{30}$$

where $\mathbf{1}$ is the unit diagonal matrix. The integral equations (23) become:

$$\boldsymbol{Z}\boldsymbol{W}_1 = \boldsymbol{\alpha}, \qquad \boldsymbol{Z}\boldsymbol{W}_2 = -\mathrm{j}\boldsymbol{\beta}, \tag{31}$$

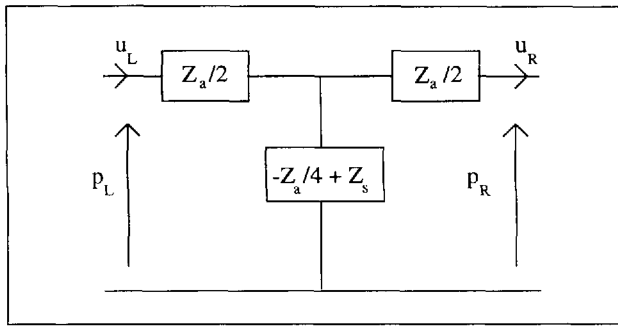

Figure 3. T-shaped circuit for the symmetrical branched tube (see equation 36).

where the matrix $\boldsymbol{Z}$ and the column vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are defined as follows:

$$\begin{aligned} \boldsymbol{Z} &= \frac{\mathrm{j}\omega\rho}{S_h^2} \int_{S_h}\int_{S_h} \varphi(\boldsymbol{r})\Big[G_h(\boldsymbol{r},\boldsymbol{r}') \\ &\quad + \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\Big]\, {}^t\varphi(\boldsymbol{r}')\,\mathrm{d}S\,\mathrm{d}S', \end{aligned} \tag{32}$$

$$\begin{aligned} \boldsymbol{\alpha} &= \frac{1}{S_h}\int_{S_h} \varphi(\boldsymbol{r})\cos(kz)\,\mathrm{d}S, \\ \boldsymbol{\beta} &= \frac{1}{S_h}\int_{S_h} \varphi(\boldsymbol{r})\sin(kz)\,\mathrm{d}S. \end{aligned} \tag{33}$$

$\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are dimensionless, $\boldsymbol{Z}$ has the dimension of an impedance, that is, the ratio of a pressure over a volume velocity.

By truncating the number of modes, equations (31) can be solved numerically. The unknown quantities T_i (equations 25) are given by the following results:

$$\begin{aligned} T_1 &= \frac{\mathrm{j}}{2} Z_c\, {}^t\boldsymbol{\beta}\boldsymbol{W}_1, & T_2 &= \frac{\mathrm{j}}{2} Z_c^2\, {}^t\boldsymbol{\beta}\boldsymbol{W}_2, \\ T_3 &= \frac{1}{2}\, {}^t\boldsymbol{\alpha}\boldsymbol{W}_1, & T_4 &= \frac{1}{2} Z_c\, {}^t\boldsymbol{\alpha}\boldsymbol{W}_2. \end{aligned} \tag{34}$$

In what follows, we restrict our study to the case where the branched tube is symmetrical with respect to the plane $z = 0$ (thus $T_1 = T_4 = 0$), and write directly the elements of the T-shaped circuit (see equations 28) shown in Figure 3, using the subscripts a and s (see equation 18) instead of 2 and 1, respectively:

$$\begin{aligned} \boldsymbol{W}_a &= \boldsymbol{W}_2, & \boldsymbol{W}_s &= \boldsymbol{W}_1, \\ Z_a &= 2T_2 = \mathrm{j}Z_c^2\, {}^t\boldsymbol{\beta}\boldsymbol{W}_a, & Z_s^{-1} &= 2T_3 = {}^t\boldsymbol{\alpha}\boldsymbol{W}_s. \end{aligned} \tag{35}$$

The modes φ_1 can be divided into symmetrical and antisymmetrical modes, so it is convenient to use submatrix notation. Because there is no interaction between these two types of modes, we write:

$$\begin{aligned} \varphi &= \begin{pmatrix} \varphi_s \\ \hline \varphi_a \end{pmatrix}, & \boldsymbol{W}_a &= \begin{pmatrix} 0 \\ \hline \boldsymbol{U}_a \end{pmatrix}, \\ \boldsymbol{W}_s &= \begin{pmatrix} \boldsymbol{U}_s \\ \hline 0 \end{pmatrix}, & \boldsymbol{Z} &= \left(\begin{array}{c|c} \boldsymbol{Z}_s & 0 \\ \hline 0 & \boldsymbol{Z}_a \end{array}\right) \\ \boldsymbol{\beta} &= \begin{pmatrix} 0 \\ \hline \boldsymbol{\beta}_a \end{pmatrix}, & \boldsymbol{\alpha} &= \begin{pmatrix} \boldsymbol{\alpha}_s \\ \hline 0 \end{pmatrix}, \end{aligned} \tag{36}$$

and rewrite equations (31), (35) and (36) in a more compact form:

$$\boldsymbol{Z}_a\boldsymbol{U}_a = -\mathrm{j}\boldsymbol{\beta}_a, \qquad \boldsymbol{Z}_s\boldsymbol{U}_s = \boldsymbol{\alpha}_s, \tag{37}$$

$$Z_a = \mathrm{j}Z_c^2\,{}^t\boldsymbol{\beta}_a\boldsymbol{U}_a, \qquad Z_s^{-1} = {}^t\boldsymbol{\alpha}_s\boldsymbol{U}_s. \tag{38}$$

Using (37), (38) can be written in two different forms:

$$Z_a = Z_c^2\,{}^t\boldsymbol{\beta}_a\boldsymbol{Z}_a^{-1}\boldsymbol{\beta}_a, \qquad Z_s^{-1} = {}^t\boldsymbol{\alpha}_s\boldsymbol{Z}_s^{-1}\boldsymbol{\alpha}_s, \tag{39}$$

$$Z_a = Z_c^2\frac{({}^t\boldsymbol{\beta}_a\boldsymbol{U}_a)^2}{{}^t\boldsymbol{U}_a\boldsymbol{Z}_a\boldsymbol{U}_a}, \qquad Z_s^{-1} = \frac{({}^t\boldsymbol{\alpha}_a\boldsymbol{U}_s)^2}{{}^t\boldsymbol{U}_s\boldsymbol{Z}_s\boldsymbol{U}_s}. \tag{40}$$

Form (39) is the most compact one, but form (40) is variational, as pointed out by Keefe [1, 2] (see also [6] for a discussion), and it can be written without modal decomposition of the velocity $v(\boldsymbol{r})$:

$$\begin{aligned} Z_a = {} & \frac{Z_c^2}{\mathrm{j}\omega\rho}\left(\int_{S_h}\sin(kz)v_a(\boldsymbol{r})\,\mathrm{d}S\right)^2 \\ & \cdot\left(\int_{S_h}\int_{S_h} v_a(\boldsymbol{r})\Big[G_h(\boldsymbol{r},\boldsymbol{r}') \right. \\ & \left. + \Re e\big\{G(\boldsymbol{r},\boldsymbol{r}')\big\}\Big]v_a(\boldsymbol{r}')\,\mathrm{d}S\,\mathrm{d}S'\right)^{-1} \end{aligned} \tag{41}$$

and

$$\begin{aligned} Z_s^{-1} = {} & \frac{1}{\mathrm{j}\omega\rho}\left(\int_{S_h}\cos(kz)v_s(\boldsymbol{r})\,\mathrm{d}S\right)^2 \\ & \cdot\left(\int_{S_h}\int_{S_h} v_s(\boldsymbol{r})\Big[G_h(\boldsymbol{r},\boldsymbol{r}') \right. \\ & \left. + \Re e\big\{G(\boldsymbol{r},\boldsymbol{r}')\big\}\Big]v_s(\boldsymbol{r}')\,\mathrm{d}S\,\mathrm{d}S'\right)^{-1}. \end{aligned} \tag{42}$$

where $v_a(\boldsymbol{r})$ and $v_s(\boldsymbol{r})$ are the two components of the velocity profile (see comments to equation 18).

3.2. Three-port representation of the junction between the main guide and a symmetrical branched tube

The above description of the effect of a side branch corresponds to a two-port representation. It could be interesting to separate the effects of the branched tube and the main waveguide in equations (32) and (39), but the shape of equations (39), involving a matrix inversion, shows that it is not possible. Nevertheless, if the side branch is long enough, that is, if its length is longer than its transverse dimensions, there is a decoupling between its input and its output: evanescent modes exist only near the extremities. It is then possible to search for a three-port representation, by distinguishing, among the symmetrical modes φ_{si}, between the planar one $\varphi_0 = 1$ and the evanescent higher order ones:

$$\boldsymbol{\varphi}_s = \begin{pmatrix}\varphi_0 \\ \hline \boldsymbol{\varphi}'_s\end{pmatrix}, \tag{43}$$

where vector $\boldsymbol{\varphi}'_s$ is obtained from $\boldsymbol{\varphi}_s$ by suppressing the planar mode. Vector $\boldsymbol{\alpha}_s$ and the matrix $\boldsymbol{Z}_s$ need to be split up similarly:

$$\boldsymbol{\alpha}_s = \begin{pmatrix}\alpha_0 \\ \hline \boldsymbol{\alpha}'_s\end{pmatrix}, \qquad \boldsymbol{Z} = \left(\begin{array}{c|c} Z_0 & {}^t\boldsymbol{z}_s \\ \hline \boldsymbol{z}_s & \boldsymbol{Z}'_s \end{array}\right). \tag{44}$$

α_0 and Z_0 are scalar quantities, $\boldsymbol{z}_s$ is a column vector, and $\boldsymbol{Z}'_s$ an infinite matrix obtained from $\boldsymbol{Z}_s$ by removing the terms depending on the planar mode. It is important to remark that integrals in $\boldsymbol{Z}_s$ involve two terms, corresponding to the function $G_h(\boldsymbol{r},\boldsymbol{r}')$ and the function $\Re e\{G_h(\boldsymbol{r},\boldsymbol{r}')\}$, respectively, (see equations 32 and 36). The first term in (32) is the input impedance matrix $\boldsymbol{Z}_h$ of the side branch, as it can be seen from equation (19) by defining:

$$\begin{aligned} p(\boldsymbol{r}) &= {}^t\boldsymbol{\varphi}(\boldsymbol{r})\boldsymbol{P}, \\ v(\boldsymbol{r}) &= \frac{1}{S_h}\,{}^t\boldsymbol{\varphi}(\boldsymbol{r})\boldsymbol{U}, \qquad \boldsymbol{P} = \boldsymbol{Z}_h\boldsymbol{U}, \end{aligned}$$

that is:

$$\boldsymbol{Z}_h = \frac{\mathrm{j}\omega\rho}{S_h^2}\int_{S_h}\int_{S_h}\boldsymbol{\varphi}(\boldsymbol{r})G_h(\boldsymbol{r},\boldsymbol{r}')\,{}^t\boldsymbol{\varphi}(\boldsymbol{r}')\,\mathrm{d}S\,\mathrm{d}S'. \tag{45}$$

If the two ends of the side branch are sufficiently far apart, then the matrix $\boldsymbol{Z}_h$ is diagonal, and the elements corresponding to the evanescent modes are equal to their respective characteristic impedances. We denote $\boldsymbol{H}$ the matrix corresponding to the second term in equation (32):

$$\boldsymbol{H} = \frac{\mathrm{j}\omega\rho}{S_h^2}\int_{S_h}\int_{S_h}\boldsymbol{\varphi}(\boldsymbol{r})\Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\,{}^t\boldsymbol{\varphi}(\boldsymbol{r})\,\mathrm{d}S\,\mathrm{d}S' \tag{46}$$

Then

$$\boldsymbol{Z}_s = \left(\begin{array}{c|c} Z_{h0} + h_0 & {}^t\boldsymbol{h}_s \\ \hline \boldsymbol{h}_s & \boldsymbol{Z}'_{hs} + \boldsymbol{H}'_s \end{array}\right). \tag{47}$$

It is possible to invert this matrix:

$$\boldsymbol{Z}_s^{-1} = \left(\begin{array}{c|c} m_0 & {}^t\boldsymbol{m} \\ \hline \boldsymbol{m} & \boldsymbol{M}' \end{array}\right), \tag{48}$$

where

$$\begin{aligned} m_0 &= 1/(Z_{h_0} + h_0 - {}^t\boldsymbol{h}_s\boldsymbol{M}''\boldsymbol{h}_s), \\ \boldsymbol{m} &= -m_0\boldsymbol{M}''\boldsymbol{h}_s, \\ \boldsymbol{M}' &= \boldsymbol{M}'' - \boldsymbol{M}''\boldsymbol{h}_s\,{}^t\boldsymbol{m} \\ \text{with}\quad \boldsymbol{M}'' &= (\boldsymbol{Z}'_{hs} + \boldsymbol{H}'_s)^{-1}. \end{aligned}$$

From equation (39), we deduce a new expression for the impedance Z_s:

$$Z_s^{-1} = \frac{n^2}{Z_{h_0} + Z_{s0}} + Y'_s, \tag{49}$$

where

$$\begin{aligned} n &= \alpha_0 - {}^t\boldsymbol{\alpha}'_s(\boldsymbol{Z}'_{hs} + \boldsymbol{H}'_s)^{-1}\boldsymbol{h}_s, \\ Z_{s0} &= h_0 - {}^t\boldsymbol{h}_s(\boldsymbol{Z}'_{hs} + \boldsymbol{H}'_s)^{-1}\boldsymbol{h}_s, \\ Y'_s &= {}^t\boldsymbol{\alpha}'_s(\boldsymbol{Z}'_{hs} + \boldsymbol{H}'_s)^{-1}\boldsymbol{\alpha}'_s. \end{aligned}$$

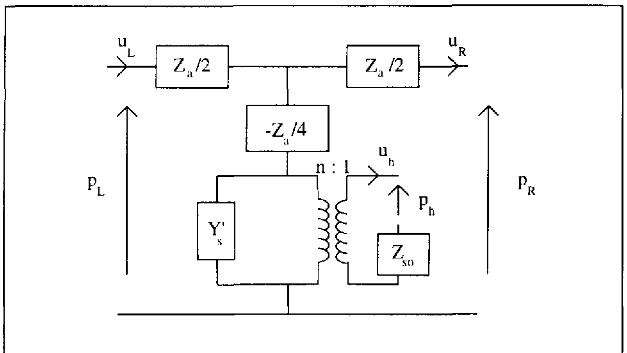

Figure 4. 3-port network for the junction between main and branched tubes (see equation 49). p_h and u_h are the planar mode components of the pressure and velocity, respectively, at the input of the branched tube, with $p_h/u_h = Z_{h_0}$ (see equation 47), n is the transformer ratio.

The corresponding equivalent circuit for the three-port network is shown in Figure 4. An important possible approximation is the plane piston approximation, and its validity is discussed below: the symmetrical higher order modes are ignored, the velocity being reduced to its planar component (in that case the antisymmetrical impedance Z_a is ignored also). One obtains:

$$Z_s^{-1} = \frac{\alpha_0^2}{Z_{h_0} + h_0}. \tag{50}$$

This value is directly obtained from the variational expression (42) by choosing a uniform profile for the velocity $v_s(\boldsymbol{r})$.

We remark that Z_{s_0} and Y'_s can be calculated by using another variation formulation, eliminating the planar mode component. As an example, with evident notations:

$$\begin{aligned} Z_{s_0} &= \mathrm{j}\omega\rho \int_{S_h}\int_{S_h} \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\,\mathrm{d}S\,\mathrm{d}S' \\ &\quad - \mathrm{j}\omega\rho \left(\int_{S_h}\int_{S_h} \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\} v'(\boldsymbol{r})\,\mathrm{d}S\,\mathrm{d}S'\right)^2 \\ &\quad \cdot \left(\int_{S_h}\int_{S_h} v'(\boldsymbol{r})\Re e\Big[\{G_h(\boldsymbol{r},\boldsymbol{r}')\} \right. \\ &\quad \left. + \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\Big] v'(\boldsymbol{r}')\,\mathrm{d}S\,\mathrm{d}S'\right)^{-1}. \end{aligned} \tag{51}$$

However, the transformer rate n cannot be simply written in such a variational form. This kind of consideration is nevertheless useful when focussing on the frequency dependence of the circuit elements in the next section.

Finally, we write the antisymmetrical field matrix $\boldsymbol{Z}_a$ in equations (39) and (40) by using the matrices $\boldsymbol{Z}_h$ and $\boldsymbol{H}$:

$$\boldsymbol{Z}_a = \boldsymbol{Z}_{ha} + \boldsymbol{H}_a, \tag{52}$$

where $\boldsymbol{Z}_{ha}$ is diagonal if the side branch is long enough (the superscript $'$ is not necessary here because no planar mode exists in the antisymmetrical field).

3.3. Low frequency limit formulae

We will analyse the nature of the elements of circuits of Figure 3 or 4 when frequency tends to zero. Detailed proofs are given in Appendix A1. If the side branch is long enough, then the input impedance of the evanescent modes is "inductive", or "acoustic masslike", and also the antisymmetrical field impedance Z_a:

$$Z_a = \mathrm{j}\omega L_a. \tag{53}$$

Concerning the symmetrical field impedance Z_s, it is written as:

$$Z_s = Z_{h_0} + Z_{s_0} \quad \text{with} \quad Z_{s_0} = \mathrm{j}\omega L_s, \tag{54}$$

where Z_{h_0} is the planar mode impedance at the input of the side branch and L_s an inertance due to the higher order symmetrical modes in this tube.

As explained by Nederveen and van Wulfften Palthe [21], the inertance L_a is negative. The inertance L_s is positive for small branched tubes and can be classically written as a "length correction" t_s for the side branch:

$$t_s = \frac{L_s S_h}{\rho}. \tag{55a}$$

Strictly speaking we notice that the expression "length correction" is not correct because it implicitly assumes a supplementary length for a tube, i.e. an "added mass" and an "added spring". As a consequence, the expression "added acoustic mass" is more convenient for L_s. Moreover, we emphasize that the length t_s is not the total length correction for the branched tube because it is necessary to take into account the quantity $-Z_a/4$ in the side branch of the circuits shown in Figures 3 and 4:

$$t_{stot} = t_s - t_a/4, \tag{55b}$$

where $t_a = L_a S_h/\rho$ (see reference [5] for a detailed study).

Nevertheless, we prefer to calculate t_s and t_a separately because these quantities correspond to the symmetrical and antisymmetrical fields, respectively. Actually, for an experimental determination using resonance frequency measurements (see [4]), these quantities are directly deduced from the even and odd resonance frequencies respectively.

Finally, the transformer rate n tends to unity when ω tends to zero, and Y'_s is of the order of ω^3. This element can be neglected in general, but Figure 4 shows that it depends on the value of the input impedance of the planar mode in the side branch, Z_{h_0}: at low frequencies this impedance is inductive (for the open tube case) or capacitive (for the closed tube case), thus Y'_s is negligible.

4. Calculation for the rectangular, 2D-case

4.1. Statement of the problem

In the case of a rectangular 2D-geometry, the transverse dimension, noted d (see Figure 5), is common to the main tube

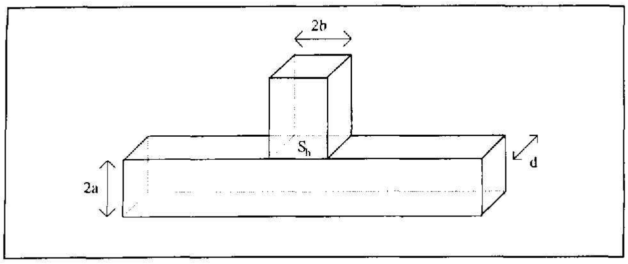

Figure 5. 2D rectangular geometry.

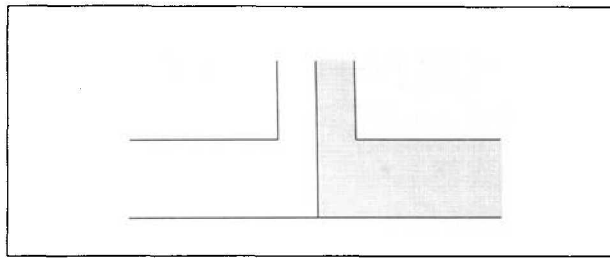

Figure 6. Equivalence between the symmetrical case and the right-angle problem.

(of width a) and the side branch (of width b). The surface S_h is planar and normal to the side branch: the condition for the modal decomposition of the last section is fulfilled, and its results can be applied.

The eigenvalues of the two waveguides and the values of the vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ and of the matrix $\boldsymbol{H}$ are given in Appendix A2.

At the limit of zero frequency, the exact results can be found by using the Schwarz-Christoffel conformal representation. Bruggeman [10] has obtained the following results, for $\delta = b/a$:

$$L_s = \frac{\rho}{\pi d}\left[\ln\left(\frac{1+\delta^2/4}{2\delta}\right) + \frac{2}{\delta}\left(1-\frac{\delta^2}{4}\right)\arctan\left(\frac{\delta}{2}\right)\right] \quad (56)$$

and

$$L_a = \frac{2\rho}{\pi d}\left[\ln\left(1+\frac{\delta^2}{4}\right) - \delta\arctan\left(\frac{\delta}{2}\right)\right]. \quad (57)$$

L_a is always negative. L_s is positive for $\delta < 1.2286$ and negative for $\delta > 1.2286$. This can be interpreted as follows: L_s corresponds to a symmetrical pressure field in the system. Thus the problem is equivalent to the problem of a right-angle in a guide (see Figure 6). In that case the total inertance L_t, including the planar mode in the main guide from $z = -b$ to $z = 0$, is related to L_s as follows:

$$L_t = 2L_s + \rho\frac{b}{2ad}.$$

If we define $b' = b/2$ and $a' = a/2$, the two waveguides play the same rôle, thus:

$$L_t(\delta') = L_t(1/\delta').$$

L_t is always positive, and L_s can be negative for higher values of δ for the following reason: the matching of the fields in the main guide at $z = 0$ is fictitious, and when L_s is deduced from L_t by subtracting the planar mode term ($\rho b/2ad$), L_s can become negative for large ratios b/a. Incidently, we have found the following property for L_s:

$$L_s\left(\frac{\delta}{2}\right) + \pi\frac{\delta}{4} = L_s\left(\frac{2}{\delta}\right) + \frac{\pi}{\delta}.$$

Thus it is possible to deduce several properties of convergence for the results obtained by modal decomposition. We summarize hereafter the results obtained by Khettabi [22].

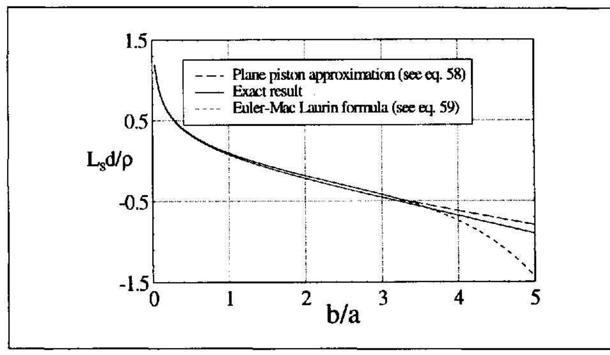

Figure 7. Series inertance as a function of the ratio $\delta = b/a$ for the 2D case: comparison between the exact result L_s, the plane piston approximation L_{sp} and the Euler-Mac Laurin formula.

4.2. Comparison with the plane piston approximation for L_s

We denote L_{sp} to be the approximate form of L_s when no higher modes are considered in the branched tube. From equations (50) and (54), the following result is obtained:

$$\frac{L_{sp}d\pi}{\rho} = -\frac{\delta\pi}{6} + \frac{2}{\delta\pi}\sum_{p\geq 1}^{\infty}\left(\frac{1}{p^2} + \frac{1}{\delta\pi}\frac{e^{-\delta p\pi}}{p^3}\right). (58)$$

Figure 7 shows the difference between this result and the exact one (equation 56). The plane piston approximation overestimates slightly the value of the inertance L_s. This kind of result can be proved by Rayleigh's method (see [17]). The relative error increases when δ increases (it is 22% for $\delta = 1$). Near $L_s = 0$, it is of course very large, then it decreases to 17% for $\delta = 5$.

We notice that it is possible to transform the series in equation (58) by using the Euler-McLaurin formula. The result is as follows:

$$\frac{L_{sp}d\pi}{\rho} = -\ln\delta - \ln\pi + \frac{3}{2} - \frac{\pi^2}{144}\delta^2 + \frac{\pi^4}{43200}\delta^4 - \frac{\pi^6}{5080200}\delta^6 + \frac{\pi^8}{435456000}\delta^8 + O(\delta^{10}) (59)$$

This approximation is excellent for $\delta < 1$ (the error being less than 0.01%). For small δ it can be compared to the expansion of the exact result:

$$\frac{L_s d\pi}{\rho} = -\ln\delta - \ln 2 + 1.$$

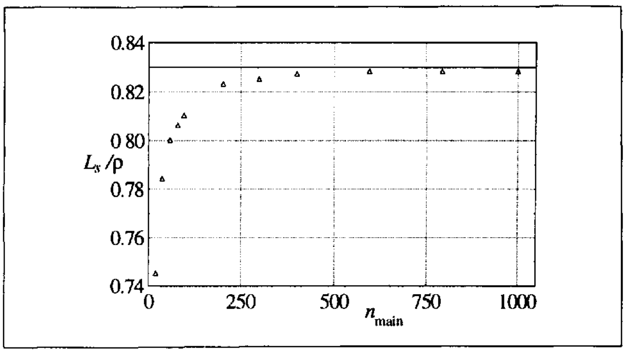

Figure 8. Convergence for the inertance L_s to the true result as a function of the number n_{main} of modes in the main guide ($\delta = 0.1$, $n_{\text{branchedtube}}/n_{\text{maintube}} = 0.1$).

The constant term is $1.5 - \ln\pi = 0.35$ for L_p and $1 - \ln 2 = 0.31$ for L_s, confirming the result $L_{sp} > L_s$.

4.3. Convergence of the result with respect to the number of modes taken into account

It is known that, for "bifurcated" electromagnetic waveguides, the ratio of the number of modes taken into account in the different guides cannot be chosen arbitrarily: Mittra [23, 24] has proven that this ratio needs to be chosen as the ratio of the heights of the guides and has made the relation between the "edge condition" at sharp angles and this is a necessity for obtaining the true result when the number of modes increases to infinity. In the present case, we have found that it is not critical to satisfy the same rule. Perhaps the convergence is faster with the true ratio, but because it is very slow in any case, it is difficult to verify this assertion.

As an example, the convergence is shown by the Figure 8, for $\delta = 0.1$ and the same ratio for the number of modes in the side branch and in the main tube. We have observed that roughly 500 modes in the main guide are necessary to obtain an error smaller than 1%. In order to improve the convergence, the following empirical law has been used (see [16]):

$$L_\infty = L_n + a_1/n^{a_2}, \tag{60}$$

where a_1 and a_2 are two constants, L_∞ is the asymptotic result and L_n is the result for the truncation at n modes in the main guide (and $n\delta$ the number of modes in the side branch). The constants a_1, a_2, L_∞ are deduced from the calculation: as an example for $\delta = 1$, and $n = 20$, 40 and 60, successively, the result is obtained with an error of 0.3%. Clearly, this is a significant improvement of the convergence. Other examples are given in [22].

4.4. Convergence of an iterative method compared to the matrix inversion

Matrices $\boldsymbol{Z}_a$ and $\boldsymbol{Z}'_s$ in equation (39) and (48), which are to be inverted, are the sum of the input impedance matrices of the side branch, $\boldsymbol{Z}_{ha}$ and $\boldsymbol{Z}'_{hs}$ respectively, and the matrices

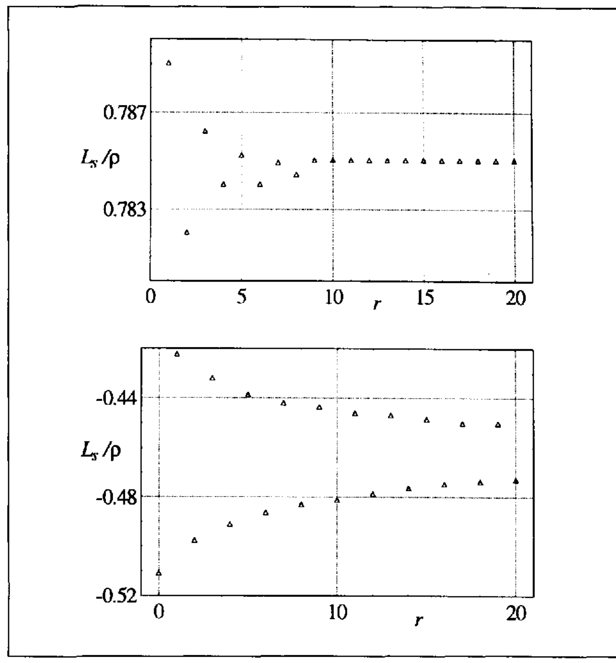

Figure 9. Convergence of the iterative calculation for the inertance L_s (see equation 61) for two ratios δ. Top: $\delta = 0.1$, $n_{\text{main}} = 40$, $n_{\text{branchedtube}} = 4$, bottom: $\delta = 3$, $n_{\text{main}} = 10$, $n_{\text{branchedtube}} = 30$, when the number of terms of the infinite series increases.

$\boldsymbol{H}_a$ and $\boldsymbol{H}'_s$ (see equation 46). It is interesting to investigate using Neumann expansion instead of matrix inversion.

Matrices $\boldsymbol{Z}_{ha}$ and $\boldsymbol{Z}'_{hs}$ are diagonal if the side branch is long enough and their norms can be assumed to be very large compared to the norms of matrices $\boldsymbol{H}_a$ and $\boldsymbol{H}'_s$, the ratios being proportional to S/S_h. We have no proof for this assumption (Kergomard [25] gave a proof for another, simpler example of discontinuity), but the numerical calculation confirms that Neumann expansion converges. The matrix to be inverted is written as follows:

$$(\boldsymbol{Z}_{ha} + \boldsymbol{H}_a)^{-1} = \boldsymbol{Y}_{ha}^{1/2}(1 + \boldsymbol{Q}_a)^{-1}\boldsymbol{Y}_{ha}^{1/2}$$

or

$$(\boldsymbol{Z}_{ha} + \boldsymbol{H}_a)^{-1} = \boldsymbol{Y}_{ha}^{1/2}\left(\sum_{r=0}^{+\infty}\left(-\boldsymbol{Q}_a\right)^r\right)\boldsymbol{Y}_{ha}^{1/2}, \tag{61}$$

where $\boldsymbol{Y}_{ha} = \boldsymbol{Z}_{ha}^{-1}$ and $\boldsymbol{Q}_a = \boldsymbol{Y}_{ha}^{1/2}\boldsymbol{H}_a\boldsymbol{Y}_{ha}^{1/2}$ (and similarly for the subscripts s).

This choice is made in order that $\boldsymbol{Q}_a$ is a symmetrical matrix. Figure 9 shows two examples of results. We will conclude with three remarks:

i. The series converges to the asymptotic result being alternately above and below the asymptotic value. The first value, obtained by truncating the series to the term $r = 0$, is the plane piston approximation. It is possible to improve the convergence by calculating an approximation of the asymptotic result:

$$L_s = \frac{L_s(0) + 3L_s(1) + 3L_s(2) + L_s(3)}{8}. \tag{62}$$

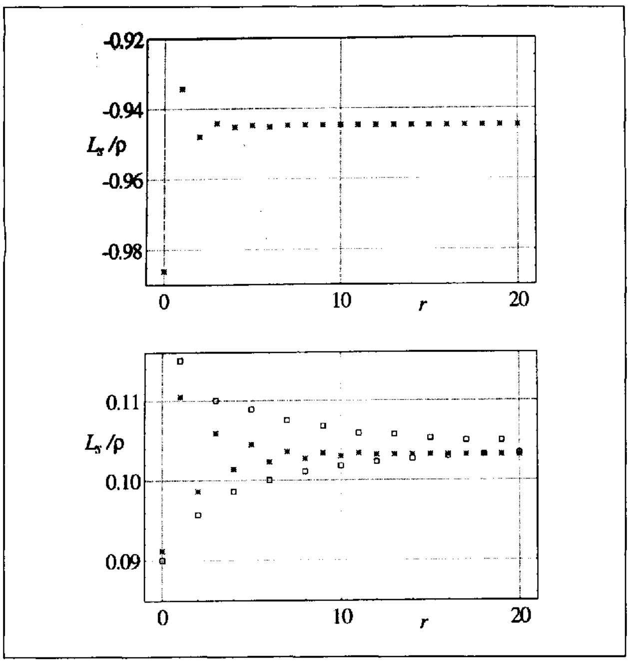

Figure 10. Acceleration of the convergence for the iterative calculation of the inertance L_s. Top: $\delta = 5$ (*) accelerated convergence; bottom: $\delta = 0.9$ (□ convergence, (*) accelerated convergence.

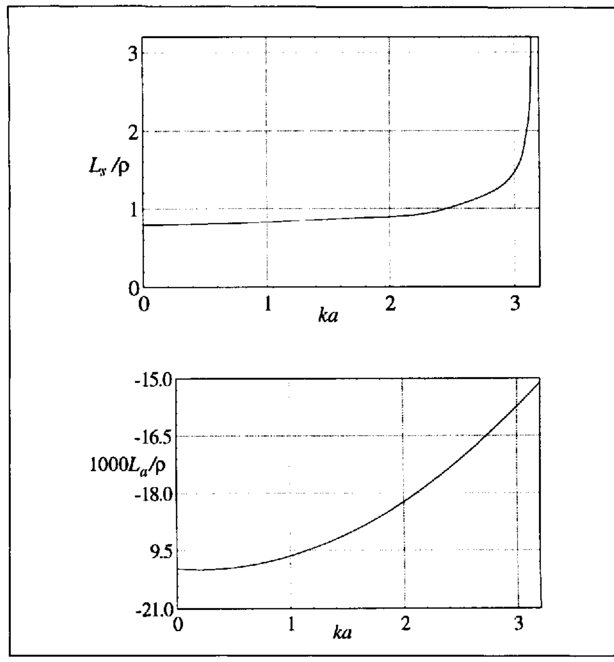

Figure 11. Variation with frequency of the inertances L_s (top, with $\delta = 0.1$) and L_a (bottom, with $\delta = 0.5$) below the first cut-off frequency of the main guide.

Here $L_s(i)$ indicates the result obtained by a truncation of the series to $r = i$.

ii. The Neumann expansion with the above approximation scheme reduces computing time. In our method, the main interest lies in the visualization of the difference from the plane piston approximation.

iii. As expected, the convergence of the Neumann expansion becomes slower when the ratio δ increases. For a large ratio ($\delta = 5$), we have found that the convergence does not exist. A solution can be found by modifying the diagonal matrix in order to increase its norm:

$$(\boldsymbol{Z}'_{hs} + \boldsymbol{H}_s)^{-1} = \left[(\boldsymbol{Z}'_{hs} + \boldsymbol{W}) + (\boldsymbol{H}_s - \boldsymbol{W})\right]^{-1}, \quad (63)$$

where $\boldsymbol{W}$ is also diagonal, and corresponds to the diagonal values of $\boldsymbol{H}'_s$ limited to the planar mode in the Green function of the main guide:

$$W_{mn} = \frac{\mathrm{j}\omega\rho}{d\pi^2}\frac{\delta}{2}\frac{\eta_m\eta_n}{n^2}\delta_{mn}.$$

Thus convergence is obtained, and for the case of small δ, it is largely improved (see Figure 10). Detailed results can be found in [22]. Similar results are obtained for the inertance L_a.

4.5. Variation with frequency below the first cutoff frequency of the main guide

As it is well known, inertances L_s and L_a vary with frequency. Figure 11 shows the result of L_s as a function of ka: we deduce from it that the variations can be neglected in practice for $ka < 1.5$, but increase rapidly for higher values. The variations of inertance L_a are smaller, as shown by Figure 11b for the same range of frequencies. Figures 11a and 11b correspond to $\delta = 0.1$ and $\delta = 0.5$, respectively.

Khettabi [22] has investigated the effect of admittance Y'_s when the frequency increases (see equation (49) or Figure 4). For resonances of the impedance $Z_{h_0} + Z_{s_0}$, the frequency shift due to this quantity cannot a priori be neglected. Nevertheless the shift is only noticeable for a long side branch when the resonance frequencies are located below the cutoff frequency of the main guide, but it can be limited by the approximate formula: (relative shift) $< 0.4b/t$, where t is the length of the side branch. As a consequence the shift is very small in practice for the frequency range for which an equivalent circuit is useful: at higher frequencies a complete calculation depending on frequency and based on equations (49) is required.

5. Numerical evaluation for the cylindrical geometry at low frequency

5.1. Specific problems of the cylindrical geometry

Consider the case of a cylindrical branched tube mounted perpendicular to a cylindrical main air column. Exact calculation of the integrals which describe the influence of the side branch discontinuity is not possible because the surface of intersection between the main waveguide and the side branch is not plane but saddle-shaped: the Green function G_h or the input impedance matrix $\boldsymbol{Z}_h$ of the side branch is not known because of the existence of the matching volume between the main air column and the side branch seen as a cylinder (see Figure 12). The surface of the main waveguide, S_h, does

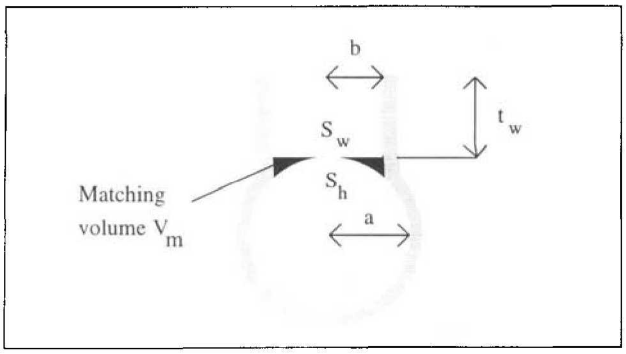

Figure 12. Cylindrical geometry. t_w is the height of the cylindrical part of the branched tube, b its radius. S_w is the surface perpendicular to the branched tube, S_h the intersection between the main guide and the branched tube.

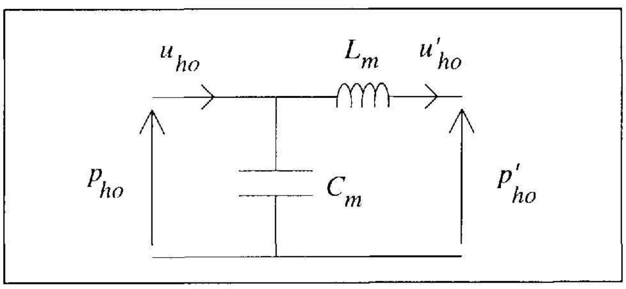

Figure 13. Equivalent circuit for the matching volume, between S_w (impedance $Z'_{h_0} = p'_{h_0}/u'_{h_0}$) and S_h (impedance $Z_{h_0} = p_{h_0}/u_{h_0}$).

not coincide with the surface S_w perpendicular to the side branch.

This matching volume is theoretically known; it depends of the radius b of the side branch and of the ratio $\delta = b/a$ of the radii of the side branch and the main waveguide [26]:

$$V_m = \frac{\pi b^3}{8}\left\{1 - \frac{4}{3\pi\delta^2}\left[(1+\delta^2)E(\delta^2) - (1-\delta^2)F(\delta^2)\right]\right\}, \tag{64}$$

where E and F are the elliptic integrals of the first and second kind, respectively, defined by [27]:

$$E(\delta^2) = \int_0^{\pi/2} \sqrt{1 - \delta^2 \sin^2\phi}\, \mathrm{d}\phi,$$

$$F(\delta^2) = \int_0^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1 - \delta^2 \sin^2\phi}}.$$

Nederveen *et al.* [5] have given a simple fit-formula, as follows:

$$\frac{t_m}{b} = \frac{\delta}{8}\left(1 + 0.207\delta^3\right).$$

As remarked by Keefe [1, 2], in the limit that the radii ratio $\delta = b/a$ is small, the excess matching volume tends to zero and the surfaces of the main waveguide and the branched tube

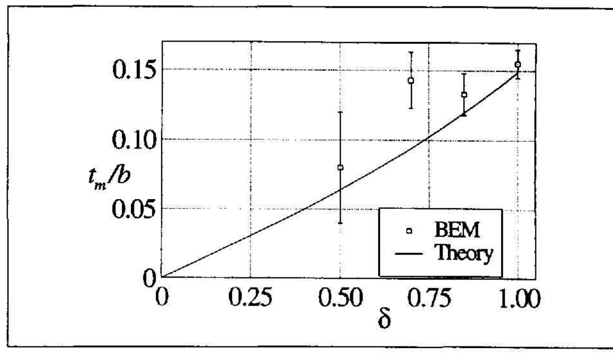

Figure 14. Equivalent length tm of the matching volume with respect to the radii ratio δ (see equation 69). —— theory, (□) obtained with Boundary Element Method.

are coincident. For the case of larger of radii ratios, the problems become more difficult. Our approximate calculation is based on the following hypotheses:

i) Symmetrical field calculation

We first use the variational formula (42) for a uniform velocity distribution over the saddle-shaped surface S_h (its corresponds to the plane piston approximation for the case of a plane surface). We are searching for the length t_s (see equation 55a and the remark of section 3.3).

From equation (A4), t_s is found to be:

$$t_s = x_0 S_h, \tag{65}$$

where

$$x_0 = \frac{1}{S_h^2} \iint_{S_h} \Re e\{G(r, r')\}\, \mathrm{d}S\, \mathrm{d}S'.$$

Then we assume that equation (A4) remains valid when Z_{h0} is the input impedance of the planar mode in the branched tube, although the input surface S_h is not planar.

The input impedance Z'_{h_0} on the plane surface S_w for an open tube is given by the following well-known relation:

$$Z'^{(o)}_{h_0} = \frac{\rho c}{S_h} \mathrm{j} \tan\left[k(t_w + t_r)\right], \tag{66}$$

where $kt_r = -\arctan(\mathrm{j}Z_r S_h/\rho c)$, Z_r being the radiation impedance, and t_w is the length of the cylindrical part of the branched tube (see Figure 12). For a closed tube, it is:

$$Z'^{(c)}_{h_0} = -\frac{\rho c}{S_h} \mathrm{j} \cot\left[kt_w\right] \tag{67}$$

(the subscripts o and c refer to the open and closed cases, respectively).

It remains to evaluate the relationship between Z_{h_0} and Z'_{h_0}. The volume between S_h and S_w being very small compared to the wavelength, one can consider that the two impedances are related by the circuit shown in Figure 13, or the following equation:

$$Z_{h_0} = \frac{Z'_{h_0} + \mathrm{j}\omega L_m}{1 + Z'_{h_0}\mathrm{j}\omega C_m}, \tag{68}$$

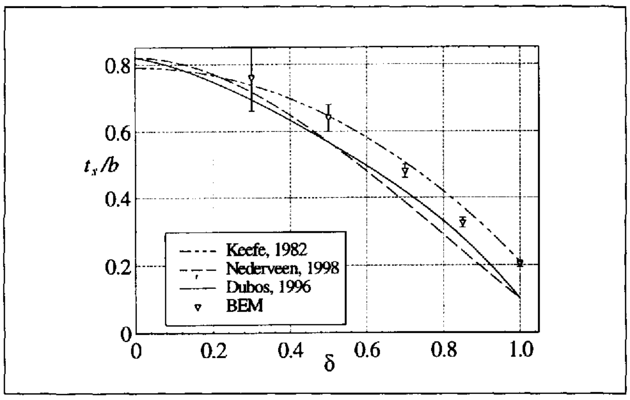

Figure 15. Inner length correction t_s due to the symmetrical modes of the branched tube versus the radii ratio $\delta = b/a$ (see equation 73).

where C_m is the compliance corresponding to the volume V_m given by equation (64):

$$C_m = \frac{V_m}{\rho c^2} = \frac{t_m S_h}{\rho c^2}, \tag{69}$$

t_m being an equivalent length for the volume shown in Figure 14 when δ varies.

Finally, the inertance L_m is unknown. For simplicity, we choose the following value:

$$L_m = \frac{\rho t_m}{S_h}. \tag{70}$$

This value is certainly overestimated because the matching volume corresponds to an enlargement of the side branch and the acoustic inertance does not satisfy the same equation as the compliance. The expression (70) allows us to obtain simple formulae for the impedance Z_{h_0} from expressions (66), (67) and (68):

$$Z_{h_0}^{(o)} = \frac{\rho c}{S_h} \mathrm{j} \tan\left[k(t_w + t_r + t_m)\right]. \tag{71}$$

For a closed tube:

$$Z_{h_0}^{(c)} = -\frac{\rho c}{S_h} \mathrm{j} \cot\left[k(t_w + t_m)\right]. \tag{72}$$

ii) Asymmetrical field calculation

We use simply formula (41) with the velocity field proportional to the coordinate z as did Keefe [1, 2]. Only the asymmetrical modes in the branched tube need to be taken into account in the Green function G_h because the integrals vanish for the symmetrical modes. The approximation here corresponds to the confusion between the two surfaces S_h and S_w.

5.2. Numerical evaluation of the elements of the electrical equivalent circuit

The different integrals involved in equations (41) and (65) have been calculated numerically for various values of δ. For the terms involving the Green function G of the main tube, the integrals on S_h have been calculated before the series, the convergence criteria for the series impose that two consecutive terms of the series are equal in the seven first digits. We present a polynomial fit for the results.

The values of the Green functions and the integrals are given in Appendix A3.

The shunt impedance Z_s due to the symmetrical modes depends on the planar mode impedance Z_{h_0} (see equations 71 and 72), the higher order modes impedance represented by the inner length correction and the radiation length in the case of an open side branch:

$$Z_s^{(c)} = \frac{\rho c}{S_h}\Big[-\mathrm{j}\cot(kt) + \mathrm{j}kt_s\Big]$$

or

$$Z_s^{(o)} = \frac{\rho c}{S_h}\Big[\mathrm{j}\tan\left[k(t+t_r)\right] + \mathrm{j}kt_s\Big], \tag{73}$$

with

$$\frac{t_s}{b} = \frac{8}{3\pi} - 0.193\delta - 1.09\delta^2 + 0.127\delta^3 - 0.71\delta^4,$$

where $t = t_w + t_m$ is the height of the side branch (including the matching volume), t_r is the radiation length correction and t_s is the equivalent length defined in equation (55a). As illustrated by Figure 15, the inner length correction t_s decreases when the diameter ratio δ increases and tends to 0.85 when this ratio tends to zero.

The computation becomes very difficult for very small δ but the limit 0.85 appears clearly: this fact confirms the validity of the modal decomposition method because the expected result is the length correction for a plane piston with an infinite flange (see e.g. [17]). Nevertheless the true expected value is that of a tube with an infinite flange, i.e. 0.82 (see reference [28]), the difference between these two values corresponding to the effect of the higher order modes in the side branch.

Finally, considering that the plane piston approximation in another case of discontinuity in circular geometry changes mainly the zeroth order term (see reference [16]), we decide to change only the first term of equation (73) into 0.82 and keep the other terms. Figure 15 compares the results with those of Keefe [1] and Nederveen *et al.* [5]: we notice that the curves given by these authors correspond to the total length correction t_{stot} (see equation 55b), but the comparison is made for t_s.

The two series inertances $Z_a/2$ symmetrically located to the left and right of the shunt admittance are found to be:

$$L_a = \rho\frac{t_a}{S}, \tag{74}$$

where

$$\frac{-t_a^{(o)}}{b\delta^2} = \Big[1.78\tanh(1.84t/b) + 0.940 + 0.540\delta + 0.285\delta^2\Big]^{-1},$$

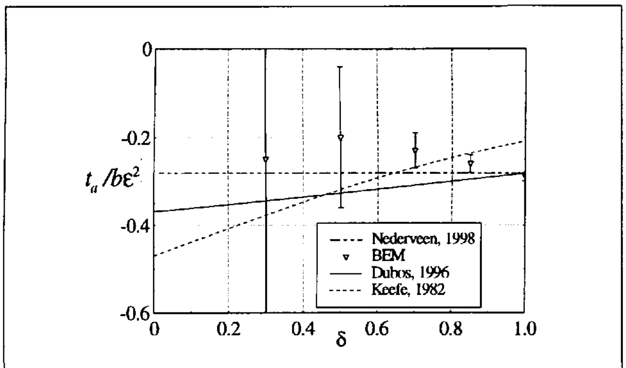

Figure 16. Equivalent length t_a due to the antisymmetrical modes of the branched tube versus the radii ratio $\delta = b/a$ (see equation 74).

and

$$\frac{-t_a^{(c)}}{b\delta^2} = \Big[1.78\coth(1.84t/b) + 0.940 + 0.540\delta + 0.285\delta^2\Big]^{-1}.$$

In this formula, as did Keefe [1, 2], we distinguish the cases of short open or closed chimneys for the Green function G_h. As a matter of fact, the Green function for a closed tube is known. For open tubes we assume that it can be calculated for a zero pressure at the open end, but it is a rough approximation: actually the calculation is very difficult, and the input impedance matrix Z_h is not diagonal.

For long chimneys ($t > b$), the results can be simplified in the following way:

$$\frac{t_a}{b\delta^2} = -0.37 + 0.087\delta.$$

All these results involve a certain number of approximations: our goal was to use an analytical calculation as far as possible. Figure 16 compares the results with those of Keefe [1] and Nederveen *et al.* [5] for large t, and Figure 17 shows the results for a given value of δ ($\delta = 1$) when the height varies.

5.3. Numerical computation using the Boundary Element Method

An independent calculation was performed using a Boundary Element Method (BEM). The geometrical modelling, meshing and acoustic modelling have been carried out using the I- DEAS Master Series software [29]. Taking into account the symmetry of the problem, the meshing is done for one side of the symmetry plane, with 6 node triangles and 8 node quadrangles (see Figure 18).

A long cylinder is considered (200 mm length) with a short side branch (10 mm) in the middle. The side branch is either "open" (zero pressure on the output surface) or closed (zero velocity on the output surface, see Figure 19). Uniform velocities are imposed at the extremities of the main tube. Symmetrical and antisymmetrical excitations at the two extremities are considered, at two frequencies: for the first one the wavelength is equal to twice the length (f = 858.424 Hz, for c = 343.3696 m/s) and for the second one equal to the length (f = 1716.848 Hz). The height of the tube (t = 10 mm) is always small compared to the wavelength.

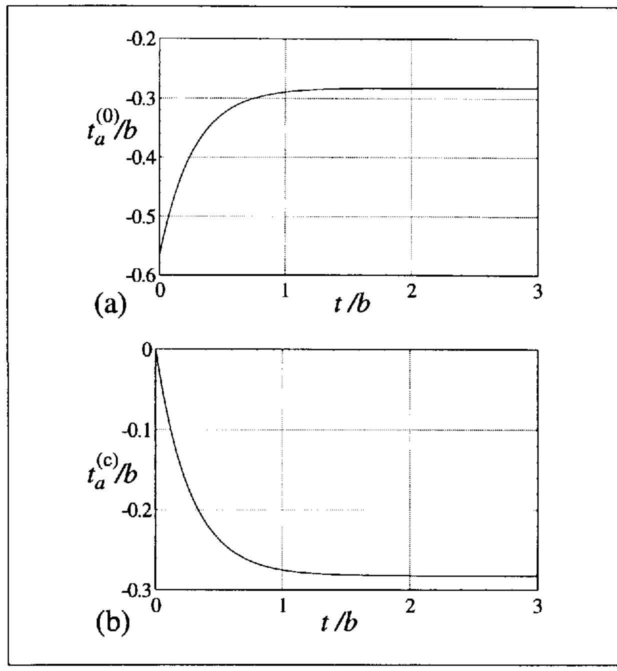

Figure 17. Equivalent length t_a ($\delta = 1$) with respect to the height of the branched tube (see equation 73). (a) case of an open branched tube (b) case of a closed branched tube.

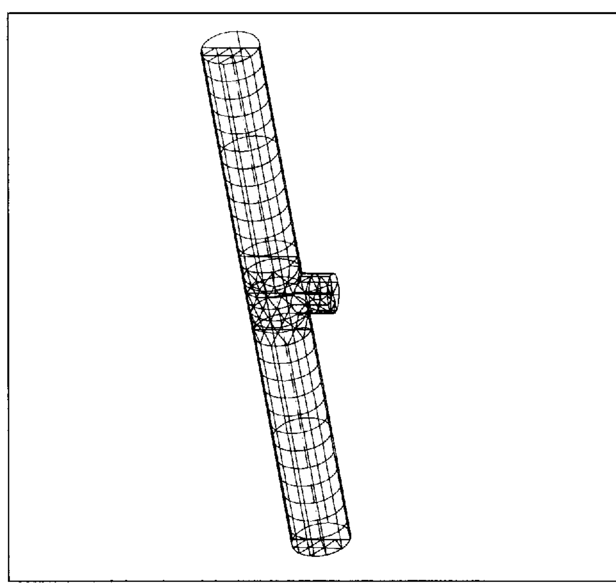

Figure 18. Meshing for the main guide and the branched tube.

In the case of a symmetrical excitation the pressure obtained at the second frequency at the end of the main tube is, for a given source velocity, proportional to the effective length of the side branch when it is open, and inversely proportional to its volume when it is closed (this quantity is an interesting test for the validity of the calculation) (see Figure 19). The series inertance is obtained with an antisym-

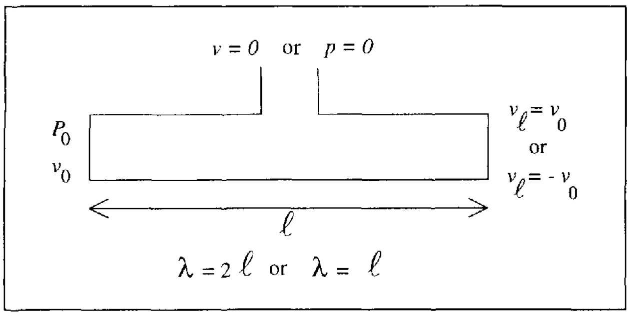

Figure 19. Schematic view of the three configurations studied with the BEM calculation: (a) $P = 0$ and $v_\ell = v_0$ with $\lambda = \ell$ give $t_s + t$; (b) $v = 0$ and $v_\ell = v_0$ with $\lambda = \ell$ give $t_m + t_s + t$; (c) $v = 0$ or $P = 0$ and $v_\ell = -v_0$ with $\lambda = \ell/2$ give t_a.

metrical excitation at the first frequency with either a closed or open tube.

The BEM modelling leads to the use of full matrices at every frequency. Nevertheless because we restrict the calculation at only two frequencies, the method is efficient. The results for the matching volume and the lengths equivalent to the shunt and series inertances L_s and L_a are shown in Figures 14, 15, 16 respectively. The uncertainties are not known but an estimation is based upon the assumption that the absolute error is the same for every case. The relative uncertainty then is larger for small diameters. For large diameters of the side branch, the results seem to be in good agreement with those obtained by modal decomposition, but the results concerning the matching volume show that the method is not very precise for small side tube diameters. For the series inertance L_s, for $\delta = 1$, the difference with modal decomposition is significant but is according to results obtained by Nederveen *et al.*.

Finally we used also the finite element method (FEM) and obtained rather satisfactory results for the length t_a but very bad results for t_s probably because in that case the volume is not strictly closed.

6. Conclusion

The modal decomposition is interesting for the calculation of sharp discontinuities with a high accuracy; nevertheless sharp discontinuities imply a great number of modes (see [16] for a discussion), and an algorithm to improve the convergence is required. We have compared exact results with the well-known plane-piston approximation.

New results have been obtained using the modal decomposition approach for the branched tube in the case of a 2D, rectangular geometry. We constructed a three-port junction representation of a branched-tube discontinuity, but our results show that it is not easy, in general, to use a variational formulation for this representation.

The treatment of the case of a cylindrical main waveguide and cylindrical branched tube has additional difficulties. On the one hand, the modal decomposition is not well adapted because the matching of the modes in the two waveguides cannot be applied over a single surface, except in the limit that the area of the branched tube is small compared to the area of the duct. On the other hand, the discretization methods lead to large uncertainties for small-diameter branched tubes, because of large discontinuities. We have proposed approximate formulae which are an accurate representation of our numerical calculations, but the inherent approximations of our numerical calculations are not completely satisfying. We note also that the re-visited calculations of one of us leads to small changes in the results.

In conclusion, these results may be useful both to the designers of musical instruments and the designers of branched-resonator silencers.

Acknowledgement

The authors acknowledge N. Joly for the BEM calculations.

Appendix

A1. Low frequency results for the general geometry

In the limit that ω approaches one, the values for the vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ in equations (33) simplify to:

$$\boldsymbol{\alpha}_s = \frac{1}{S_h} \int \varphi_s(\boldsymbol{r})\,\mathrm{d}S,$$

thus $\alpha_0 = 1$ and $\boldsymbol{\alpha}'_s = \mathbf{0}$ ($\boldsymbol{\alpha}'_s$ is of the order ω^2).

$$\boldsymbol{\beta}_s = \frac{k}{S_h} \int \varphi_a(\boldsymbol{r})\,\mathrm{d}S. \tag{A1}$$

From equation (5), the real part of the Green function for the main tube becomes:

$$G(\boldsymbol{r},\boldsymbol{r}') = -\frac{|z-z'|}{2S} + \sum_{\substack{m,n>0\\ m+n\neq 0}}^{\infty} \frac{\mathrm{e}^{-\eta_{mn}|z-z'|}}{2\eta_{mn}S}\psi_{mn}(\boldsymbol{w})\psi_{mn}(\boldsymbol{w}'). \tag{A2}$$

It can be deduced that in the low frequency limit, α is independent of ω, and β and $\boldsymbol{H}$ (see equation 46) are proportional to ω. If the side branch is long enough, the input impedance of the evanescent modes depends on the frequency as $\mathrm{j}\omega$ (it is *inductive*). Finally, using equations (39) and (52), the element Z_a is found to be an inertance L_a (or acoustic mass) written as:

$$\frac{L_a}{\rho} = -\frac{1}{S^2}\,{}^t\tilde{\boldsymbol{\beta}}_a \boldsymbol{X}_a^{-1} \tilde{\boldsymbol{\beta}}_a$$

with

$$\tilde{\boldsymbol{\beta}}_a = \frac{1}{S_h} \int_{S_h} z\varphi_a\,\mathrm{d}S, \tag{A3}$$

and

$$\boldsymbol{X}_a = \frac{1}{S_h^2}\int_{S_h}\int_{S_h}\varphi_a(\boldsymbol{r})\Re e\{G(\boldsymbol{r},\boldsymbol{r}')\} \cdot {}^t\varphi_a(\boldsymbol{r}')\,\mathrm{d}S\,\mathrm{d}S' + \frac{\boldsymbol{Z}_{ha}}{\mathrm{j}\omega\rho}.$$

The second term of $\boldsymbol{X}_a$ is diagonal, each element being proportional to the eigenvalue η_{hai} of the modes in the branched tube:

$$\left(\frac{\boldsymbol{Z}_{ha}}{\mathrm{j}\omega\rho}\right)_{ij} = \eta_{hai}^{-1}\delta_{ij}.$$

Similarly, from equation (49), an inertance L_s corresponding to the rôle of the symmetrical modes in the branched tube can be found as follows:

$$Z_s = Z_{h_0} + \mathrm{j}\omega L_s, \tag{A4}$$

where

$$L_s = \rho\left[x_0 - {}^t\boldsymbol{x}_s\boldsymbol{X}_s^{-1}\boldsymbol{x}_s\right],$$

with

$$x_0 = \frac{1}{S_h^2}\int_{S_h}\int_{S_h}\Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\,\mathrm{d}S\,\mathrm{d}S' = \frac{h_0}{\mathrm{j}\omega\rho},$$

$$\boldsymbol{x}_s = \frac{1}{S_h^2}\int_{S_h}\int_{S_h}\varphi_s\Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\,\mathrm{d}S\,\mathrm{d}S' = \frac{\boldsymbol{h}_s}{\mathrm{j}\omega\rho},$$

$$\boldsymbol{X}_s = \frac{1}{S_h^2}\int_{S_h}\int_{S_h}\varphi_s\Re e\{G(\boldsymbol{r},\boldsymbol{r}')\} \cdot {}^t\varphi_s\,\mathrm{d}S\,\mathrm{d}S' + \frac{\boldsymbol{Z}_{hs}}{\mathrm{j}\omega\rho}.$$

The plane piston approximation L_p for L_s is given by: $L_p = \rho x_0$.

A2. Vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, and matrix $\boldsymbol{H}$ for the 2D-geometry

Khettabi [22] has found the following values for the vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ for the 2D-geometry:

$$\boldsymbol{\alpha}_i = \gamma_i\frac{kb\sin(kb)}{(kb)^2 - (\eta_{hi}b)^2},$$

$$\boldsymbol{\beta}_i = \mathrm{j}\sqrt{2}\frac{kb\cos(kb)}{(kb)^2 - (\eta_{hi}b)^2},$$

with $\eta_{hi} = i\pi/2b$ and $\gamma_i = \sqrt{2-\delta_{0i}}$, δ_{0i} being the Kronecker symbol.

The matrix $\boldsymbol{H}$ is expressed by:

$$H_{ij} = \frac{\mathrm{j}\omega\rho}{2d\pi^2}\delta\gamma_i\gamma_j\sum_{p=0}^{\infty}\frac{\gamma_p^2}{j^2 - q_p^2\delta^2} \cdot \left[\delta_{ij} + \delta_{i(-j)} + \frac{2X_p}{\pi(i^2 - q_p^2)\delta}\right],$$

with:

$$q_p = \sqrt{p^2 - \left(\frac{2ka}{\pi}\right)^2},$$

$$X_0 = \frac{2ka}{\pi}(-1)^i\sin(2kb),$$

$$X_p = q_p\left[(-1)^i\mathrm{e}^{-\pi\delta q_p} - 1\right], \quad p \neq 0.$$

A3. Numerical computation in cylindrical geometry at low frequencies

The Green function of the infinite main guide is given by equation (5). The eigenmodes and eigenvalues in cylindrical geometry are written as:

$$\psi_{mn}(\boldsymbol{w}) = \cos(m\theta)\frac{J_m(\mu_{mn}u/b)}{J_m(\mu_{mn})}\frac{\sqrt{2}}{\sqrt{1-(m/\mu_{mn})^2}},$$

$$k_{mn} = \frac{-\mathrm{j}\mu_{mn}}{a\sqrt{1-(ka/\mu_{mn})^2}},$$

where k is the plane mode eigenvalue, $\boldsymbol{w} = (u,\theta)$, a is the radius of the main guide, J_m is the Bessel function of order m, μ_{mn} is the n-th zero of the Bessel derivative.

At the input of a closed branched tube, the Green function can be written as:

$$G_h(\boldsymbol{w}_b, 0; \boldsymbol{w}_b', 0) = -\frac{\cot(kt)}{kS_h} + \frac{1}{S_h}\sum_{m,n\neq 0}\frac{\psi_{mn}(\boldsymbol{w}_b)\psi_{mn}(\boldsymbol{w}_b')}{\alpha_{mn}}\coth(\alpha_{mn}t),$$

where $t = t_w + t_m$ is the height of the branched tube (including the matching volume), $\boldsymbol{w}_b$ is the two-dimensional coordinate in the transverse section of the branched tube, and $\alpha_{mn} = k_{mn}$.

At the input of an open side branch, the Green function can be written as:

$$G_h(\boldsymbol{w}_b, 0; \boldsymbol{w}_b', 0) = \frac{\tan k(t+t_r)}{kS_h} + \frac{1}{S_h}\sum_{m,n\neq 0}\frac{\psi_{mn}(\boldsymbol{w}_b)\psi_{mn}(\boldsymbol{w}_b')}{\alpha_{mn}}\tanh(\alpha_{mn}t),$$

where radiation effects on the higher order modes are ignored.

If the branched tube is sufficiently long ($t > 2b$), the hyperbolic functions tend to unity and the input impedance of the higher order modes is their characteristic impedance, independently of the output condition.

From Appendix A1, using the particular choice for the velocity profiles, the equivalent length t_s is computed from equation (65). The series length correction ta is found to be:

$$t_a = S_h\left(\int_{S_h} z^2\,\mathrm{d}S\right)^2 \cdot\left(\int_{S_h}\int_{S_h} z\Big[G_h(\boldsymbol{r},\boldsymbol{r}') + \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\Big] z'\,\mathrm{d}S\,\mathrm{d}S'\right)^{-1}.$$

Therefore five integrals over the surface S_h of intersection between the branched tube and the main guide need to be calculated:

$$I_1 = \int_{S_h}\int_{S_h} \Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}\,\mathrm{d}S\,\mathrm{d}S',$$
$$J_1 = \int_{S_h} z^2\,\mathrm{d}S,$$
$$J_2 = \int_{S_h}\int_{S_h} zG_h^{(c)}(\boldsymbol{r},\boldsymbol{r}')z'\,\mathrm{d}S\,\mathrm{d}S',$$
$$J_3 = \int_{S_h}\int_{S_h} zG_h^{(o)}(\boldsymbol{r},\boldsymbol{r}')z'\,\mathrm{d}S\,\mathrm{d}S',$$
$$J_4 = \int_{S_h}\int_{S_h} z\Re e\{G(\boldsymbol{r},\boldsymbol{r}')\}z'\,\mathrm{d}S\,\mathrm{d}S'.$$

The subscript (o) refers to the open branched tube, the subscript (c) refers to the closed branched tube. The integral J_1 can be written as:

$$J_1 = 2ab^3k\int_{-1}^{1} x^2 \arcsin\left(\delta\sqrt{1-x^2}\right)\mathrm{d}x,$$

where $\delta = b/a$.

The quantities J_2 and J_3 are written as:

$$J_2 = 2\pi b^5\sum_{n=0}^{\infty}\frac{\coth(\mu_{1n}t/b)}{\mu_{1n}^3[1-(1/\mu_{mn})^2]}\left(\frac{J_2(\mu_{1n})}{J_1(\mu_{1n})}\right)^2,$$

$$J_3 = 2\pi b^5\sum_{n=0}^{\infty}\frac{\tanh(\mu_{1n}t/b)}{\mu_{1n}^3[1-(1/\mu_{mn})^2]}\left(\frac{J_2(\mu_{1n})}{J_1(\mu_{1n})}\right)^2.$$

A great simplification occurs if only the first term of the series ($n=0$) is taken into account. Actually, if we note

$$f(\mu_{1n}) = \frac{1}{\mu_{1n}^3[1-(1/\mu_{1n})^2]}\left(\frac{J_2(\mu_{1n})}{J_1(\mu_{1n})}\right)^2,$$

we can see that the terms corresponding to $n \geq 1$ are negligible compared to the first term $n=0$:

$$\frac{f(\mu_{11})}{f(\mu_{10})} = 0.0036 \quad\text{and}\quad \frac{f(\mu_{12})}{f(\mu_{10})} = 0.00033.$$

The quantities I_1 and J_4, depending on the real part of the Green function of the main guide, can be decomposed in three parts: one part containing only the planar mode, a second part containing a single sum over the modes ($m=0, n\geq 1$) and a third part containing a double sum over the modes ($m \geq 1$, $n \geq 0$):

$$I_1 = I_1^p + I_1^{e_1} + I_1^{e_2},$$
$$J_4 = J_4^p + J_4^{e_1} + J_4^{e_2},$$

where:

$$I_1^p = -\frac{2b^3}{\pi}\int_{-1}^{1}\int_{-1}^{1}\arcsin\left(\delta\sqrt{1-x^2}\right)\cdot\arcsin\left(\delta\sqrt{1-y^2}\right)|x-y|\,\mathrm{d}x\,\mathrm{d}y,$$

$$I_1^{e_1} = \frac{2ab^2}{\pi}\sum_{n=1}^{\infty}\frac{1}{\mu_{0n}}\int_{-1}^{1}\int_{-1}^{1}\arcsin\left(\delta\sqrt{1-x^2}\right)\cdot\arcsin\left(\delta\sqrt{1-y^2}\right)\mathrm{e}^{-\mu_{0n}\delta|x-y|}\,\mathrm{d}x\,\mathrm{d}y,$$

$$I_1^{e_2} = \frac{4a^3\delta^2}{\pi}\sum_{m=1}^{\infty}\frac{1}{m^2}\sum_{n=0}^{\infty}\frac{1}{\mu_{mn}[1-(1/\mu_{mn})^2]}\cdot\int_{-1}^{1}\int_{-1}^{1}\sin\left[m\arcsin\left(\delta\sqrt{1-x^2}\right)\right]\cdot\sin\left[m\arcsin\left(\delta\sqrt{1-y^2}\right)\right]\cdot\mathrm{e}^{-\mu_{mn}\delta|x-y|}\,\mathrm{d}x\,\mathrm{d}y,$$

$$J_4^p = -\frac{2b^5}{\pi}\int_{-1}^{1}\int_{-1}^{1}\arcsin\left(\delta\sqrt{1-x^2}\right)\cdot\arcsin\left(\delta\sqrt{1-y^2}\right)xy|x-y|\,\mathrm{d}x\,\mathrm{d}y,$$

$$J_4^{e_1} = \frac{2ab^4}{\pi}\sum_{n=1}^{\infty}\frac{1}{\mu_{0n}}\int_{-1}^{1}\int_{-1}^{1}\arcsin\left(\delta\sqrt{1-x^2}\right)\cdot\arcsin\left(\delta\sqrt{1-y^2}\right)xy\mathrm{e}^{-\mu_{0n}\delta|x-y|}\,\mathrm{d}x\,\mathrm{d}y,$$

$$J_4^{e_2} = \frac{4a^5\delta^2}{\pi}\sum_{m=1}^{\infty}\frac{1}{m^2}\sum_{n=0}^{\infty}\frac{1}{\mu_{mn}[1-(m/\mu_{mn})^2]}\cdot\int_{-1}^{1}\int_{-1}^{1}\sin\left[m\arcsin\left(\delta\sqrt{1-x^2}\right)\right]\cdot\sin\left[m\arcsin\left(\delta\sqrt{1-y^2}\right)\right]\cdot xy\mathrm{e}^{-\mu_{mn}\delta|x-y|}\,\mathrm{d}x\,\mathrm{d}y.$$

All integrals have been calculated numerically (FORTRAN programming in double precision, IMSL library) as a function of the ratio δ; for the case of the quantities $I_1^{e_1}$, $I_1^{e_2}$, $J_4^{e_1}$, $J_4^{e_2}$, the integral have been calculated first, and then the convergence of the series has been studied. The convergence criteria have been chosen as: two successive terms of the series are equal in the seven first digits. We have noticed that the smaller is the value of δ, the higher is the number of modes needed for the convergence (for this criteria, up to 10000 terms have been taken into account to obtain the convergence of the series).

Finally, the numerical values for the equivalent lengths have been calculated as functions of δ using a polynomial interpolation.

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